

Partial Differential Equations

Problem Sheet 1

1. Consider the Cauchy problem

$$\begin{cases} u_x = 0, & (x, y) \in \Omega, \\ u(x, x^2) = g(x), & x \in \mathbb{R}, \end{cases}$$

where

$$\Omega = \{(x, y) \in \mathbb{R}^2 : y > 0\}.$$

- (a) Determine the necessary and sufficient condition on the function g for the existence of a solution.
 - (b) If a solution exists, show that it is unique.
 - (c) Suppose instead that $\Omega = \mathbb{R}^2$. Does the Cauchy problem admit a solution? If so, is it unique?
2. For each $n \in \mathbb{N}$, verify that the function

$$\phi_n(x, y) = \begin{cases} y \cos(ny), & y < 0, \\ y \cos y, & y \geq 0, \end{cases}$$

is a solution of the following Cauchy problem:

$$\begin{cases} u_x = 0, & \text{in } \Omega = \mathbb{R}^2, \\ u(x, y) = x^2 \cos y, & (x, y) \in \Gamma, \end{cases}$$

where $\Gamma = \{(x, y) \in \mathbb{R}^2 : y = x^2\}$. (Relate this with Part(c) of question 1.)

3. Consider the Cauchy problem

$$\begin{cases} u_y = 0, & (x, y) \in \Omega, \\ u(x, y) = e^x \cos y, & (x, y) \in \Gamma, \end{cases}$$

where

$$\Gamma = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

- (a) Show that the problem admits a unique solution when

$$\Omega = \{(x, y) \in \mathbb{R}^2 : |x| \leq 1\}.$$

- (b) Determine whether the problem admits a solution when $\Omega = \mathbb{R}^2$. If a solution exists, is it unique?